

Chemistry — Assignment 4: Functional Group Identification

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Chemistry (Foundational Chemistry) · Semester 5

Question. Identify the functional group(s) present in each given molecule, then count the number of different functional groups in the final multi-group molecule.

Reference groups.

- | | |
|--|---|
| • alcohol: -OH | • amide: -CONH_2 |
| • ether: -O- | • nitrile: $\text{-C}\equiv\text{N}$ |
| • aldehyde: -CHO | • amine: -NH_2 |
| • ketone: >C=O | • thiol: -SH |
| • carboxylic acid: -COOH | • nitro: -NO_2 |
| • ester: -COOR | • halide: -X |
| • acid chloride: -COCl | |

Worked identifications

(i) -SH **thiol**, -OH **alcohol**, C=O **ketone/aldehyde**, -COCl **acid chloride**.

(ii) $\text{-C}\equiv\text{N}$ **nitrile**, -OH **alcohol**, C=O **ketone**, $\text{N-H } 2^\circ$ **amine**.

(iii) -CHO **aldehyde**, -SH **thiol**, -O- **ether**, C=O **ketone**.

Count: identify every characteristic group, then count the *distinct types* $\rightarrow$ the final molecule contains **5 different functional groups** (ring amine, ether -OCH_3 , primary amine -NH_2 , nitrile/ring-N, and C-Cl halide).